# TraceSmith — 系统架构 · GPU Trace 端到端数据流

Application → bindings → capture / state / format → replay & export → GPU backends

Primary flow
Fallback / feedback
Dashed = architecture zone
Box = module + responsibility

**PROJECT OVERVIEW** TraceSmith is a cross-platform GPU profiling and deterministic replay system for AI compiler, framework, and driver engineers.
Its C++17 core captures low-overhead execution events, stores compact SBT traces, and exports timelines for analysis and debugging.

## APPLICATION · 用户入口

**C++ / Python Application**
嵌入业务进程

**CLI**
profile · record · view · export

**External UI**
Perfetto · Chrome

## BINDINGS · Python / C++ 边界

**_tracesmith.so**
pybind11 native extension

## CORE LIBRARY · C++17

**Capture Module**
CUPTIProfiler · MCPTIProfiler
Metal · Ascend · ROCm · eBPF

**Common + Format**
RingBuffer SPSC · StackCapture
SBTWriter / SBTReader

**State Module**
GPUStateMachine
TimelineBuilder · PerfettoExporter

**Replay Module**
ReplayEngine · StreamScheduler
DeterminismChecker · FrameCapture

## CROSS-CUTTING · 集群与调试

replay / debug feedback

**Cluster Module**
GPUTopology · TimeSync
NCCLTracker

**GDB / Tracy**
RSPHandler · GPUDebugEngine
TracyExporter

## GPU BACKENDS · 系统接口

**NVIDIA**
CUDA · CUPTI · nsys

**MetaX**
MACA · MCPTI · mcTracer

**Apple / AMD**
Metal · xctrace · ROCm

**Huawei / Linux**
CANN · msprof · eBPF

## OUTPUTS · Trace 产物

**trace.sbt**
compact binary

**trace.json / .perfetto**
Perfetto timeline

**graph.dot / ASCII**
analysis views

主链：GPU event → lock-free ring → SBT/state → replay 或 Perfetto；Cluster / GDB 为旁路能力。